AQUABOTS — Document Updates (Session: 2026-04-22)

Purpose: Capture updates to project docs following water test and planning session. Since project files are read-only in this workspace, this document contains the exact text to merge into each affected file.

Affected documents:

- `AQUABOTS_POWER_SYSTEM_CHANGELOG.md` — append new Rev entry, update header
- `AQUABOTS_WIRING_GUIDE.md` — Rev bump WG-3 → WG-4 (after tomorrow's install)
- `AQUABOTS_EMI_FILTERING_BOM.pdf` — deviation notes pending Rev ISR-5 regeneration
- `SOFTWARE_INTEGRATION_ACTION_ITEM.md` — Phase 2 propulsion status update

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1. AQUABOTS_POWER_SYSTEM_CHANGELOG.md

1.1 Update header (line 5)

Change:

```
**Last Updated:** 2026-03-04 (Rev ISR-4)
```

To:

```
**Last Updated:** 2026-04-22 (Rev ISR-5 planned for 2026-04-23)
```

1.2 Append new changelog entry (insert at line 177, above Rev ISR-4 heading)

```
### Rev ISR-5 (2026-04-23) — First Water Test + Servo Swap + EMI Install

**Water test 2026-04-22 (first in-hull test):**
- AUTO mode: boat executed CCW circles, mission aborted.
- Manual RC: could only turn left with forward throttle (per Robert).
- Rudder observed centered on retrieval — non-diagnostic (abort triggers HOLD = c
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- Servo used: 3.5 kg·cm interim unit (production DSServo delayed due to BEC integ
- Secondary issue: RC link dropping after several minutes of use.

**Root-cause hypotheses (ranked):**
1. Insufficient servo torque against propeller wash + rudder too small / misposit
2. SERVO1_TRIM off-center (not yet ruled out at time of water test).
3. Compass calibration stale post-hull-close — could cause auto-mode heading drif
4. EMI from motor into uncalibrated compass — motor EMF on compass was not filter

**Planned remediation 2026-04-23:**
1. Bench diagnostic on interim servo to rule out SERVO1_TRIM / SERVO1_MIN / SERVO
2. Swap to production DSServo. Wire new BEC per P15/P16 branch spec.
3. Full EMI filter install — all components on hand, none previously installed.
4. QGC recalibration: SERVO1 parameters, compass, accelerometer. Verify SERVO1_FU
5. Symmetric travel check targeting ≥25°/side, capped at ~35° to avoid hydrodynam
6. RC dropout investigation — check new BEC rail sag under servo load, verify fai

**Field procedures (printouts generated 2026-04-22):**
- `STEERING_TROUBLESHOOT_AND_SERVO_SWAP.pdf`
- `EMI_FILTERING_INSTALL_CHECKLIST.pdf`

**BOM deviations flagged for Rev ISR-5 regeneration:**
- F5 ferrite bead count: BOM says 6+ (4 I2C + 2 UART optional); actual install =
- Copper tape targets: BOM doesn't list ESC body, BEC body, or buck converter bod

**Outstanding after this rev:**
- Production DSServo installed + calibrated → log final SERVO1 param values.
- EMI install verification: compass drift under motor load <2° target.
- AUTO-mode water test retry (only after compass recal post-EMI install).
```

1.3 Update “Section 5: KNOWN ISSUES & FUTURE WORK” (at line 382)

Add entries under Known Issues:

- `[NEW 2026-04-22]` Rudder authority insufficient with 3.5 kg·cm interim servo — CCW circling observed. Production DSServo + larger rudder (Kama) in progress.
- `[NEW 2026-04-22]` RC link dropping after several minutes — root cause TBD. Hypotheses: BEC rail sag, RX antenna routing, power module interaction. Will test with new BEC tomorrow.

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2. AQUABOTS_WIRING_GUIDE.md — Rev WG-3 → WG-4 (planned)

Do NOT bump the rev until physical install on 2026-04-23 is complete. After that, update:

2.1 Header rev bump

Bump to `Rev WG-4 (2026-04-23)` with summary: "EMI filtering installed. Production DSServo + new BEC wired. Compass + accel recalibrated post-install."

2.2 Copper tape table additions (in section 4 or wherever copper tape is enumerated)

Add new entries:

Target	Type	Drain Point	Notes
ESC body	Component body wrap	(–) busbar via short wire	Avoid heatsink fins — do not thermally trap
BEC body	Component body wrap	(–) busbar	Over existing shrink-wrap
5V buck converter body	Component body wrap	(–) busbar	—
12V buck converter body	Component body wrap	(–) busbar	—

2.3 Document steering servo parameters

Add to Section 3 (or equivalent configuration section):

```
## Pixhawk Steering Servo Configuration (SERVO1)

| Parameter | Value | Notes |
|---|---|---|
| SERVO1_FUNCTION | 26 | GroundSteering (ArduRover) |
| SERVO1_MIN | [fill after install] | Match to mechanical left stop |
| SERVO1_MAX | [fill after install] | Match to mechanical right stop |
| SERVO1_TRIM | [fill after install] | PWM value that produces mechanically-cente
| SERVO1_REVERSED | [0 or 1] | Set per bench test (left stick = rudder port side)

**Travel target:** ≥25° deflection each side of center. Cap at ~35° to avoid hydr
**Servo model (production):** DSServo [model TBD — record part number on install]
**Power:** Via Castle BEC on P15/P16 branch (ATO 2A SB).
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3. AQUABOTS_EMI_FILTERING_BOM — Rev ISR-5 regen notes

This is a PDF; content source-of-truth lives elsewhere. When regenerating to Rev ISR-5:

3.1 Correct F5 row

Current: `F5 Axial Ferrite Bead (thru-hole) ~70 Ohm @ 100MHz, 3.5x6mm 6+ I2C SDA/SCL (4), opt UART (2)`

Should read: `F5 Axial Ferrite Bead (thru-hole) ~70 Ohm @ 100MHz, 3.5x6mm 2 S8 pH I2C SDA + SCL only (IMU S7 removed WG-2)`

3.2 Add copper tape (SH1) placement rows

Existing SH1 row only references cable shielding. Add component-body targets:

Target	Type	Drain
ESC body	Component wrap	(–) busbar
BEC body	Component wrap	(–) busbar
5V buck body	Component wrap	(–) busbar
12V buck body	Component wrap	(–) busbar

3.3 Status column updates (after install 2026-04-23)

All `ORDER` entries → `INSTALLED` with install date footnote.

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4. SOFTWARE_INTEGRATION_ACTION_ITEM.md — Phase 2 update

4.1 Phase 2.1 Propulsion Interface Layer

Change status from 🔲 `NOT STARTED` to 🟡 `IN PROGRESS` and add:

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**Status update 2026-04-22:**
- ESC calibrated via QGC ✅
- SERVO3 (throttle) parameter baseline established ✅
- SERVO1 (steering) — bench-verified in manual RC but insufficient rudder authori
- First water test completed — identified servo torque + rudder sizing issue, not
```

```
- ArduPilot Rover MANUAL mode control path verified end-to-end (RC → Pixhawk → ES
```

4.2 Dependencies update

Under "Current Blockers" change: 🔴 `**Propulsion hardware integration:** Waiting on thruster mounting & ESC setup`

To: 🟡 `**Propulsion hardware integration:** Mechanical rudder authority insufficient with interim servo. Production servo + larger rudder installing 2026-04-23. Software path verified.`

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5. Cross-reference — What This Session Changed

Item	Before	After
Water test status	Planned	First test complete (2026-04-22)
Servo	Interim 3.5 kg·cm	Production DSServo (install 2026-04-23)
BEC	Interim wiring	New BEC per P15/P16 spec (install 2026-04-23)
EMI filters	Parts on hand, none installed	Full install (2026-04-23)
Rudder	Current size	Larger (Kama working on redesign)
SERVO1 params	Not documented	To be documented post-calibration
RC link	Reliable	Dropping after minutes — investigation pending

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6. Follow-ups Not Yet in Project Docs

These are worth surfacing into a dedicated issues list somewhere:

1. RC link dropout — hypothesize new BEC alleviates, but verify with bench soak test before next water test.

2. Coordinate with Kama on new rudder dimensions, mounting plan, and whether linkage geometry changes.
3. Confirm with Robert that the PM07-style power module removal does not eliminate battery V/I telemetry to Pixhawk in ArduPilot (open item from earlier).
4. Jetson sensor parsing — still unresolved whether any Jetson-side code handles the current `TEMP:x,SAL:y,PH:z` format. Claude Code discovery task handed off previously.
5. Compass cal must be redone AGAIN after EMI install — expect heading drift to drop significantly once filters are in.